



USL Assignment 3

Implement Singular Value Decomposition algorithm

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SVD THEORY

1. Introduction to Singular Value Decomposition

Singular Value Decomposition (SVD) is a powerful mathematical technique used in linear algebra and machine learning for matrix factorization and dimensionality reduction. It decomposes a matrix into three separate matrices that capture the essential structure of the original data.

SVD is widely used in data science to simplify complex datasets by reducing the number of features while preserving the most important information. This helps improve computational efficiency and allows machine learning models to focus on the most meaningful patterns in the data.



Unlike some dimensionality reduction methods that rely on class labels, SVD is an unsupervised learning technique, meaning it analyzes the structure of the dataset without using target labels.

2. Mathematical Concept Behind LDA

For any matrix A of size $m \times n$, Singular Value Decomposition represents the matrix as:

$$A = U \Sigma V^T$$

Where:

- U is an orthogonal matrix containing the left singular vectors.
- Σ (**Sigma**) is a diagonal matrix containing singular values.
- V^T is the transpose of the matrix containing the right singular vectors.

The singular values in the diagonal matrix represent the importance of each corresponding component. Larger singular values indicate components that capture more information about the dataset.

By selecting only the top **k singular values**, it is possible to approximate the original dataset with fewer dimensions while retaining most of its important structure.



3. Steps of the SVD Algorithm

The main steps involved in applying Singular Value Decomposition are:

- Standardize the dataset to ensure features have comparable scales
- Represent the dataset as a numerical matrix
- Compute the Singular Value Decomposition of the matrix
- Identify the singular values that capture the most variance
- Select the top **k components** corresponding to the largest singular values
- Transform the dataset into a lower-dimensional feature space

This process reduces the dimensionality of the dataset while preserving its most significant information.

4. Example of SVD

Consider a simple matrix representing a dataset:

$$A = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix}$$

Using Singular Value Decomposition, this matrix can be decomposed into three matrices:

$$A = U \Sigma V^T$$

Where:



- $\mathbf{U}$ contains the left singular vectors
- $\mathbf{\Sigma}$ contains the singular values in descending order
- $\mathbf{V}^T$ contains the right singular vectors

The singular values indicate how much information each component contributes to representing the original dataset.

By keeping only the largest singular values, the dataset can be represented using fewer dimensions while maintaining most of the important information.

5. SVD Applied to Olivetti Faces Dataset

The **Olivetti Faces dataset** consists of grayscale facial images of multiple individuals. Each image contains thousands of pixel values that represent the intensity of different regions of the face.

SVD is applied to:

- Identify the most important patterns within the facial images
- Reduce the dimensionality of the dataset
- Extract meaningful feature representations from high-dimensional image data



After applying SVD, the dataset is transformed into a lower-dimensional feature space where each component captures important structural information from the original images.

The reduced dataset is then used to train a classification model and compare the performance before and after applying dimensionality reduction.

6. Inferences from SVD Output

After applying Singular Value Decomposition to the dataset, the explained variance ratio indicates how much information is retained by each singular component.

The cumulative explained variance plot helps determine the optimal number of components required to represent most of the dataset information.

In this experiment, dimensionality reduction using SVD significantly reduced the number of features while preserving most of the important information contained in the original dataset.

The classification accuracy obtained using the reduced dataset was comparable to the accuracy obtained using the full dataset, indicating that SVD successfully eliminated redundant features while retaining meaningful patterns in the data.



7. References

<https://scikit-learn.org/stable/modules/generated/sklearn.decomposition.TruncatedSVD.html>

<https://www.geeksforgeeks.org/data-science/singular-value-decomposition-svd/>

<https://numpy.org/doc/stable/reference/generated/numpy.linalg.svd.html>